

Santosh Guntuka

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SUMMARY

Entry-level Software Engineer with practical experience in backend development, RESTful API design, and Agile methodologies. Adept at integrating systems, automating processes, and enhancing performance through rigorous testing and streamlined CI/CD practices. Proficient in Python with a commitment to collaboration and continuous process improvement across multidisciplinary teams.

SKILLS

- **Programming Languages:** CSS, HTML, JavaScript, Python, SQL, TypeScript, Java
- **Web Development Frameworks:** React, Next.js, Express, Node.js, GraphQL
- **Cloud & DevOps Tools:** AWS, Docker, Kubernetes, Git
- **Databases:** MongoDB, PostgreSQL
- **Machine Learning:** TensorFlow, PyTorch, PySyft

EXPERIENCE

Forefront Consulting

Feb 2025 - Present

Software Developer

San Diego, CA

- Designed and integrated RESTful APIs, leveraging PostgreSQL for efficient data management in scalable backend systems.
- Developed and rigorously tested core features using Python and Django, ensuring clear, error-free code that met system performance targets.
- Implemented secure authentication with JWT and enhanced deployment workflows through CI/CD automation with Jenkins.
- Collaborated effectively in an Agile team environment by actively participating in sprint planning, code reviews, and daily stand-ups.

Dartmouth and Hitchcock

Apr 2024 - Feb 2025

Machine Learning Researcher

Lebanon, New Hampshire

- Developed and automated federated learning workflows using Python, improving deployment efficiency and ensuring well-documented, testable code.
- Implemented PySyft-based federated learning setups to enable secure, distributed model training while adhering to rigorous testing practices.
- Configured and containerized federated learning environments with Docker, ensuring scalability and reproducibility with clear code documentation.
- Evaluated and tested various federated learning tools and frameworks, conducting thorough performance and security assessments.

California State University, Dominguez Hills

Mar 2023 - Dec 2023

IT Service Management

Los Angeles, CA

- Maintained and supported Windows Active Directory, handling user account management, group policies, and security configurations.
- Diagnosed and troubleshooting Windows and macOS systems, addressing OS, hardware, and networking issues to ensure system stability and performance.
- Managed and resolved Tier 2 support tickets via ServiceNow, achieving a 90% resolution rate within 24 hours, improving end-user experience.
- Provided technical support for enterprise applications, VPN configurations, and system access, enhancing operational efficiency.

California State University, Dominguez Hills

Jun 2022 - Feb 2023

Graduate Research Assistant

Los Angeles, CA

- Analyzed RL convergence, improving algorithm accuracy by 25% and reducing convergence time by 20% through detailed benchmarking.
- Developed and executed simulations using Python and Amazon SageMaker, optimizing RL model training and performance evaluation.
- Achieved a 30% reduction in computational time while maintaining a 90% success rate in finding feasible paths under probabilistic constraints.
- Benchmarked various Reinforcement Learning approaches, documented algorithm performance, and fine-tuned hyperparameters to ensure optimal solution convergence.

EDUCATION

California State University, Dominguez Hills | Master of Science, Computer Science (GPA: 3.7)

2023

PROJECTS

Chess Engine

- Developed a chess AI engine using Python, Pygame, and TensorFlow to play autonomously and predict moves based on large chess datasets.
- Designed and implemented data preprocessing pipelines for efficient PGN file processing and incremental training of the neural network.
- Optimized model performance through advanced neural network architectures, dropout, and batch processing techniques.
- Enhanced the Pygame UI for real-time chessboard updates, intuitive user interaction, and visual improvements.

Genetics SNP data simulation: Predicting a binary outcome in simulated genomic data

Collaboration with Cedars-Sinai Medical Center

Los Angeles, CA

- Contributed to a Cedars Sinai Medical Center research project, applying predictive modeling to genomic datasets to uncover complex genetic patterns.
- Analyzed algorithm effectiveness in genomic data, demonstrating a 15% improvement in prediction accuracy with customized techniques, advancing genomic research.